

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Habitat case studies: woodland, rainforest, reefs, islands, mangroves, Antarctica - Set 1

Mark scheme

AQA spec 3.1.2.3.4; The living environment; textbook pp. 50-71

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • No major temperature extremes; regular water supply with no pronounced dry season; deep fertile soils built by leaf litter and root binding; strong seasonality (1) • Deciduous canopy present spring to autumn shades the floor during the best growing temperatures (1) • Bluebells, snowdrops, lesser celandine grow and flower early in spring before the trees leaf (1) • Winter food shortage causes animals to hibernate (dormice), store food or migrate (1)	4
01.2	<i>1 mark for each point.</i> • Long time for species to colonise; some woodland plants colonise very slowly so their presence (ancient woodland indicators) shows the wood is old (1) • Natural woodland has trees of all ages with clearings where old trees died, giving varied light and dead wood habitats (1) • A new plantation has a single-age dense canopy with low light and few niches until trees start to die (over 200 years) (1)	3
01.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Clearance for farmland, plantations, urban development, roads, mineral extraction (1) • Fragmentation isolating remaining woods so populations cannot move between them and are vulnerable (1) • Management change: end of coppicing so woods become uniform and dark; replacement by conifer monocultures (1) • Overgrazing by deer preventing regeneration; introduced pathogens (ash dieback) (1)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	
02.1	<i>Up to 5 marks in total; award marks in proportion to the completeness of each point.</i> • A: coppice rotation creates a patchwork of different ages / light levels - clearings with wildflowers for butterflies, dense regrowth for nesting birds; oak supports many invertebrates (up to 2) • B: same tree species but the closed canopy shades the floor so few flowering plants, fewer invertebrates and butterflies; fewer niches (up to 2) • C: non-indigenous single-species, single-age conifers support few native invertebrates; dense shade; no clearings or dead wood; needles acidify soil (up to 2) • Habitat diversity / structural variety increases the range of niches and hence species (up to 2)	5
02.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Designated protected areas: SSSIs, NNRs (Sherwood Forest), SACs (Epping Forest), National Parks (1) • Legal protection of ancient woodland: planning permission refused for developments destroying irreplaceable habitats (2012 NPPF) (1) • Conservation management: coppicing, creating clearings, planting mixed native species, leaving dead wood, controlling deer (1) • Planting new native woodlands linking fragments (corridors); Woodland Trust / National Trust ownership (1)	3
02.3	<i>1 mark for each point.</i> • No species is completely dominant and food webs have many alternative links (1) • So a change in the population of one species has a relatively small impact on the whole community (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	
03.1	<i>1 mark for each point.</i> • Stable tropical climate over long periods gave time for species to evolve and adapt to local abiotic and biotic factors; many rare species with small ranges (1) • High light, warm temperatures and high rainfall all year allow continuous photosynthesis and growth, supporting rich food webs with no seasonal food shortage (no hibernation / migration) (1) • Stratification (emergent, canopy, understorey, shrub, ground layers) and epiphytes create many niches; intense competition for light drives specialisation (1) • Biotic pressures (predation, inter-species relationships) drive evolution of specialised pollination / seed dispersal relationships; plants flower and fruit at different times (1)	4
03.2	<i>1 mark for each point.</i> • Adapted to stable conditions so they have narrow ranges of tolerance; a small change can move conditions outside the range (1) • Many species are rare with small ranges / small populations (1)	2

Question	Marking guidance	Marks
03.3	<i>1 mark for each point.</i> • No cold winter kills herbivores so plants defend themselves with toxic chemicals (alkaloids) that are medicinally active in controlled doses (quinine, diosgenin, curare) (1) • Animals without teeth / claws (frogs, reptiles) rely on chemical defences or venoms (fer-de-lance venom - captopril) (1)	2
03.4	<i>1 mark for each point.</i> • Over half the Amazon's rainfall comes from transpiration within the forest; clearance reduces downwind rainfall (1) • Rapid decay recycles nutrients so soils are shallow / poor; canopy and roots prevent erosion of the thin soil - clearance leads to rapid erosion and infertility (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 52-55; AQA spec 3.1.2.3.4.	
04.1	<i>1 mark for each point.</i> • Small clearings were farmed for a few years then abandoned as nutrients declined; the forest recolonised and soil fertility recovered before the area was cleared again (1) • Population growth shortens the gap between clearances so the forest and soil do not recover (1) • Clearings enlarge and degrade; less regrowth (1)	3
04.2	<i>1 mark for each point.</i> • Rainforest soils are thin and nutrient-poor (nutrients held in the biomass); once cleared, fertility falls in a few years and yields drop, so more land is cleared (1) • Abandoned land may not be recolonised if no forest survives nearby; erosion of exposed soil (1)	2
04.3	<i>1 mark for each point.</i> • Mining (aluminium, iron, gold, copper) destroys habitat directly and pollutes rivers with turbid drainage, cyanide and mercury from processing (1) • Reservoirs (eg Tucurui, 2850 km²) flood large areas, block migratory fish and river dolphins; decomposing vegetation produces acidic, deoxygenated water; fluctuating downstream flows (1) • Droughts increase frequency, area and duration of fires (long fires boil sap and kill trees); cloud forests lose humidity as clouds form higher so frogs and epiphytes decline; some models predict replacement by savannah (1)	3
04.4	<i>1 mark for each point.</i> • A creditor country or conservation charity cancels / buys part of a debtor country's foreign debt (1) • In return the country guarantees to protect an area of rainforest (eg USA - Guatemala, Brazil; WWF - Costa Rica, Philippines) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 55-57; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
05.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Protected areas: Korup NP Cameroon (1260 km², 600 tree species, endemic <i>Ancistrocladus korupensis</i> with anti-HIV / antimalarial alkaloids); Tumucumaque NP Brazil (39 000 km², adjoining French Guiana reserve); Alto Maues; Rio Bravo Belize funded by charities, companies and ecotourism • Debt for nature swaps; sustainable exploitation: selective logging of high-value timber (mahogany) funding conservation; ITTO aims (largely unmet); CITES listing of timber species (Honduras mahogany Appendix II) • Ecotourism income (but visitor damage in Costa Rica, Malaysia); community projects and employment; carbon sequestration payments • Effectiveness: protected areas work where enforced but face illegal logging, mining, poaching, roads, conflict; drivers (population growth, demand for beef, soya, palm oil, timber, minerals) originate outside the forest; climate change cannot be controlled locally • Judgement: combination of large protected areas, economic alternatives for local people and reduced international demand needed; overall loss continues	9
05.2	<i>1 mark.</i> • Carbon sequestration reducing climate change; genetic resources / medicines; rainfall regulation; global biodiversity (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 56-57; AQA spec 3.1.2.3.4.	
06.1	<i>1 mark for each point.</i> • Zooxanthellae are photosynthetic algae living inside polyp cells (up to 30 000 per mm²) (1) • They use the polyp's CO₂ and excretory wastes (nitrogen, phosphorus) for photosynthesis and growth, so nutrients are recycled efficiently (1) • They supply the polyp with glucose and oxygen - over 95% of its food energy; the polyp provides a protected habitat and nutrients (1) • Polyps also trap plankton with cilia and stinging nematocysts; surplus CO₂ is used to build the calcium carbonate skeleton (1)	4
06.2	<i>1 mark for each point.</i> • High light (shallow, clear water) for the algae; warm stable temperatures (about 18-30 °C); low turbidity; constant salinity; well-oxygenated water (1) • Found only where all these coincide (about 1% of ocean area) (1) • Small changes (warming, sediment, freshwater runoff) move conditions outside the narrow range of tolerance (1)	3

Question	Marking guidance	Marks
06.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Fisheries: support over a quarter of marine species including fish, lobsters, conch (1) • Erosion protection: complex structure dissipates wave energy protecting low coral islands; coral sand builds shores (1) • Medicinal discoveries: AZT and Ara-C from sponges, dolastatin from sea hares, bryostatin from Bugula, trabectedin from sea squirts, secosteroids, coral for bone grafts (1) • Climate control: carbonate skeletons sequester CO₂ (1) • Tourism (diving, snorkelling) (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-60; AQA spec 3.1.2.3.4.	
07.1	<i>1 mark for each point.</i> • Overall decline from 62% to 22% (1) • Sharp falls in 1998 (to 35%) and 2016 (to 18%) coinciding with bleaching events, with partial recovery after each (1)	2
07.2	<i>1 mark for each point.</i> • The polyps expel their symbiotic zooxanthellae so the coral loses colour; bleached corals are alive but grow very slowly and may die if algae do not return (1) • Raised water temperature (climate change / El Niño) (1) • Pollution: sewage, pesticides, cyanide, sunscreen; low oxygen from zooplankton blooms after overfishing (1) • High light / solar activity; ocean acidification hinders skeleton formation (1)	4
07.3	<i>1 mark for each point.</i> • Coastal construction and dredging increase sedimentation / turbidity covering and killing polyps (1) • Sewage and fertiliser nutrients cause algal growth that shades corals and prevents egg release (1) • Physical damage from anchors, divers, souvenir collection; loss of mangroves / seagrass nurseries that trap sediment (1)	3
07.4	<i>1 mark.</i> • Protected breeding populations grow larger and produce surplus young / adults that spill over into fished areas (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 60-63; AQA spec 3.1.2.3.4.	
08.1	<i>1 mark for each point.</i> • Giant tritons are predators of the crown-of-thorns starfish which eats coral polyps (1) • Removing tritons allowed starfish populations to explode, destroying large areas of reef (food web / trophic cascade) (1)	2

Question	Marking guidance	Marks
08.2	<i>1 mark for each point.</i> • Fixed mooring buoys so boats do not anchor on coral (1) • Education of divers / snorkellers not to touch coral or kick up sediment; bans on spear fishing and souvenir collection (1) • Fishing controls: no-take zones (Chagos, Great Barrier Reef, Lundy, Lamlash Bay), seasons, minimum and maximum catch sizes, bans on dynamite / small-mesh nets / trawling (1) • Protection of turtle nesting beaches; control of coastal development; protection of mangroves and seagrass; artificial reefs (reef balls, sunk ships / oil rigs) (1)	4
08.3	<i>1 mark for each point.</i> • No sunlight so no symbiotic algae; polyps rely only on captured food so have far less energy and grow very slowly (1) • Damage from trawls takes hundreds of years to repair; the slow-growing, low-fecundity fish (orange roughy) they support are easily over-exploited (1)	2
08.4	<i>1 mark for each point.</i> • Deep-water trawling (also by-catch); oil and gas exploration in deeper water; ocean acidification (1) • Protected areas banning trawling, eg Darwin Mounds SAC off NW Scotland, Oculina Bank off Florida (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 61-64; AQA spec 3.1.2.3.4.	